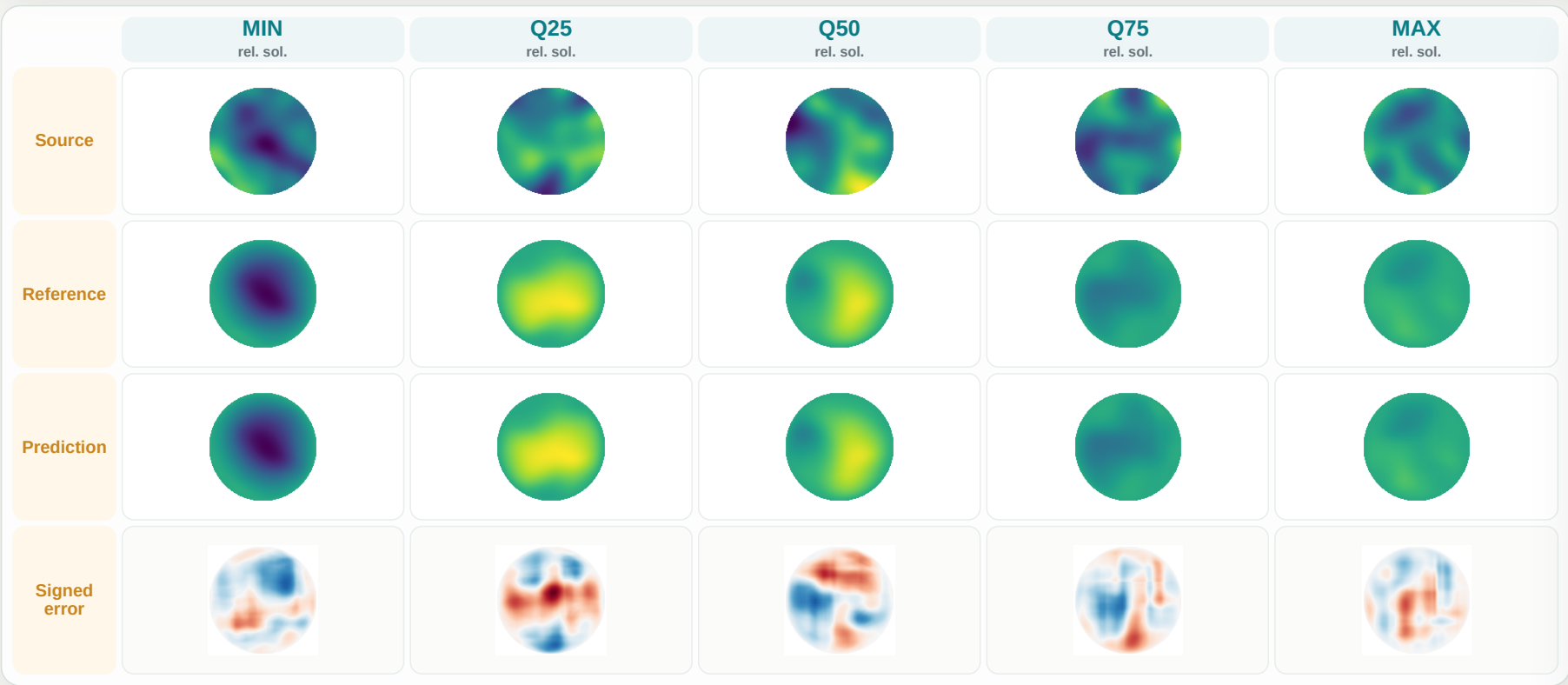


Numerical Evidence II: CouplingNet Solution Reconstruction

Quantile-selected samples show source, reference, prediction, and signed error

$$\Omega = \{x^2 + y^2 < 0.5^2\}, \quad -\nabla \cdot (a \nabla u) = f, \quad a(x, y) = 1 + \frac{1}{2} \sin(2\pi x) \sin(4\pi y), \quad \mathbf{b} = 0, \quad c = 0.$$



SELECTED BY RELATIVE SOLUTION ERROR

CASE	REL. SOL.	SPLIT ENERGY
min	1.87%	7.48e-4
q25	3.10%	8.59e-4
q50	3.95%	7.41e-4
q75	5.33%	6.87e-4
max	8.84%	4.77e-4

Reference and prediction share one solution scale; the error row uses a zero-centered signed scale.

The learned directional source split reconstructs the 2D solution across the relative-error distribution, not only for a single selected case.